Assignment 8: Fitting a Model to Stock Data

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1 Introduction

Financial markets generate large volumes of sequential data that can be analyzed using time-series modeling techniques. Understanding historical price behavior can provide insights into market dynamics and risk characteristics, even though predicting short-term stock movements remains challenging.

In this study, historical stock data for Apple Inc. (AAPL) is analyzed using a statistical time-series model. The objective is to fit an ARIMA model to daily stock returns and evaluate its predictive performance. In addition to prediction accuracy, key financial performance indicators are computed to evaluate the stock's risk-adjusted returns.

Three financial metrics are considered in this analysis: Annual Return, Sharpe Ratio, and Sortino Ratio. These indicators are widely used in quantitative finance to evaluate investment performance while accounting for volatility and downside risk. By combining statistical modeling with financial evaluation metrics, this analysis demonstrates a complete workflow for financial time-series analysis.

2 Dataset and Preprocessing

Historical stock price data was obtained from Yahoo Finance using the `yfinance` Python package. The dataset contains approximately five years of daily trading data for Apple Inc., including information such as opening price, closing price, trading volume, and adjusted closing price.

For this analysis, the adjusted closing price was selected because it accounts for stock splits and dividend adjustments, making it more suitable for long-term analysis. From this series, daily returns were calculated using the following formula:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where P_t represents the adjusted closing price at time t. The resulting return series represents the percentage change in price from one trading day to the next.

Non-trading days such as weekends and market holidays introduce gaps in the raw dataset. These observations were removed during preprocessing to ensure a continuous return series suitable for modeling.

Figure 1 illustrates the adjusted closing price of AAPL during the observation period, while Figure 2 shows the daily return series derived from the price data.

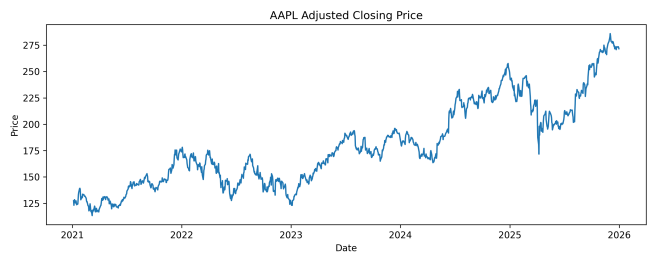

Figure 1: Adjusted closing price of AAPL over the observation period.

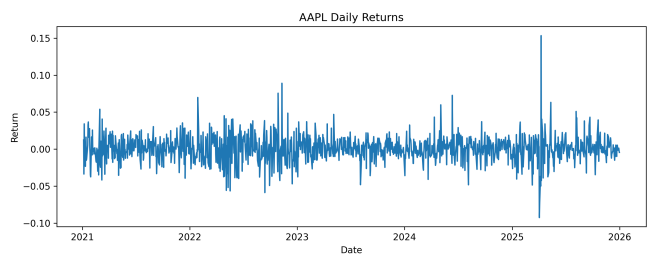

Figure 2: Daily returns derived from adjusted closing prices.

3 Model Description

To model the temporal behavior of stock returns, an Autoregressive Integrated Moving Average (ARIMA) model was used. ARIMA models are widely applied in financial time-series analysis because they can capture autocorrelation patterns present in sequential data.

The ARIMA model is defined by three parameters:

$$ARIMA(p, d, q)$$

where p represents the number of autoregressive terms, d represents the degree of differencing applied to the data, and q represents the number of moving-average terms.

Because financial return series are typically stationary, no differencing was required in this case ($d = 0$). An ARIMA(2,0,2) model was selected to capture short-term dependencies in the return series.

The dataset was split chronologically to preserve the temporal ordering of the observations. The first 80% of the dataset was used for training the model, while the remaining 20% was reserved as unseen test data. The trained model was then used to generate forecasts for the test period.

Figure 3 shows the predicted daily returns compared with the observed returns during the evaluation period.

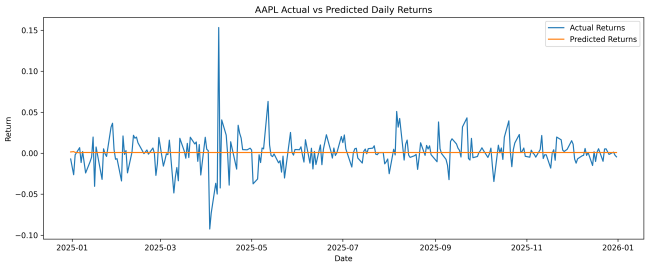

Figure 3: Comparison between predicted and actual daily returns.

4 Model Evaluation

The predictive performance of the model was evaluated using two common regression metrics: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). These metrics measure the average difference between predicted and actual values.

Metric	Value
RMSE	0.0204
MAE	0.0128

Table 1: Prediction error metrics for the ARIMA model.

The results indicate that the prediction error is relatively large compared to the magnitude of daily returns. This behavior is expected because financial returns contain substantial randomness and are difficult to forecast accurately in the short term.

5 Financial Metrics

To evaluate the investment performance of the stock during the unseen test period, three financial indicators were calculated.

Annual Return measures the compounded yearly return of the asset. It represents the overall growth rate of an investment over time.

Sharpe Ratio evaluates the return generated per unit of volatility. It is defined as:

$$Sharpe = \frac{\mu_r}{\sigma_r}\sqrt{252}$$

where μ_r is the mean daily return and σ_r is the standard deviation of returns.

Sortino Ratio is similar to the Sharpe Ratio but considers only downside volatility, penalizing negative returns rather than total variability.

Metric	Value
Annual Return	0.0832
Sharpe Ratio	0.4058
Sortino Ratio	0.5571

Table 2: Financial performance metrics during the test period.

Figure 4 shows the cumulative growth of a $1 investment during the evaluation period.

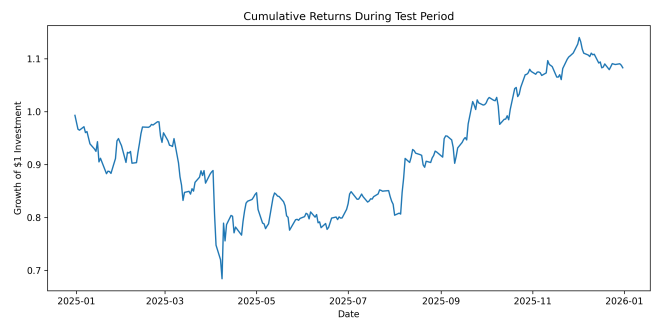

Figure 4: Cumulative growth of $1 investment during the test period.

The results indicate a positive annual return during the evaluation period. However, the relatively modest Sharpe and Sortino ratios suggest that volatility played an important role in the observed returns.

6 Conclusion

This analysis demonstrated a complete workflow for financial time-series modeling. Historical stock data for Apple Inc. was collected, preprocessed, and used to train an ARIMA model on daily returns. The model's predictive performance was evaluated using RMSE and MAE, while financial metrics were computed to assess risk-adjusted performance.

Although the model was able to capture limited short-term patterns, prediction accuracy remained constrained by the inherent randomness of financial markets. These findings highlight the challenges associated with forecasting stock returns and emphasize the importance of evaluating investments using risk-adjusted performance metrics.